HELPS: Auditable Screening of Potential Helicopter Landing Areas From Dated Aerial Imagery

Research manuscript draft

Abstract—Preflight screening of unprepared terrain can use dated imagery to narrow the search for possible helicopter landing areas. We present the Helicopter Emergency Landing Place Suggestion System (HELPS), a local workflow that converts a georeferenced RGB image and a user-supplied boundary into a reviewable terrain package. The workflow composes overlapping Segment Anything ViT-H masks into regions, classifies them using an existing random forest, and joins predicted low vegetation and bare or sparse ground across tile boundaries. It retains valid 30 m slope cells at or below 15 degrees and applies geometric gates before a reviewer can select the package for aircraft-specific research assessment. On a dated 512 m Raichur crop, a historical bare-ground-only producer yielded no qualifying polygons despite identifying 6646.75 square metres of possible bare ground. A separate 128 m single-tile prototype run with the two-class seed policy yielded two qualifying terrain footprints. These runs demonstrate empty and nonempty screening outcomes. The differing footprints and rules, the absence of independent reference labels, and the coarse elevation data prevent comparative accuracy or landing-suitability claims.

Index Terms—Aerial imagery, geographic information systems, helicopter landing areas, random forests, remote sensing, terrain screening.

I. INTRODUCTION

Unprepared helicopter landing areas may vary in usable extent, surface condition, slope, nearby obstructions, and access. Reconnaissance and aircraft-specific procedures remain necessary for operational decisions [1]. HELPS addresses the earlier map-based research task of finding terrain footprints that warrant examination. A candidate is a polygon supported by the specified terrain and geometry gates. A research opportunity is a candidate or slope subregion evaluated for a particular aircraft and mission context.

The engineering contribution connects a dated RGB observation to the selected regional terrain package used for assessment. A common input contract and tiled processing associate the classified output with an image and date. A terrain audit records how slope coverage and geometry affect the candidates. The review step separates generating a package from selecting it for assessment. We report historical and prototype evidence of feasibility and define a controlled evaluation of image scale and seed policy.

II. RELATED WORK

Scherer *et al.* assessed site geometry and approach feasibility with aircraft-mounted lidar and demonstrated autonomous landing on a full-scale helicopter [2]. Their system measured hazards near the time of landing. HELPS uses an offline dated observation and exposes unknown evidence to the reviewer. Erskine *et al.* compared modelled helicopter landing areas with surveyed footprints using land-cover and elevation inputs from 1 to 30 m. They found that feature scale and site geography affected boundary accuracy [3]. The independent reference survey in that study is an important distinction from the feasibility evidence reported here.

Segment Anything (SAM) provides promptable and automatic mask generation [4]. SamGeo and related remote-sensing studies demonstrate georeferenced use of SAM [5] [6]. HELPS reuses the project-specific GeoSAM region composition and trained random forest from the Bhadra project [7], based on Breiman's ensemble method [8]. The forest's semantic labels and the downstream terrain gates are distinct from SAM mask quality.

III. DATA AND STUDY AREAS

Raichur imagery and elevation

WELL Labs contributed the drone orthomosaic *RAICHUR_D10_2025_09-M1* to OpenAerialMap [9] [10]. The archived catalog metadata lists an interval from 18:30 to 19:30 UTC on 31 August 2025, corresponding to 00:00–01:00 India Standard Time on 1 September 2025. We use that local capture date. The nominal 3.1 cm source was resampled to a 0.5 m RGB grid in WGS 84 / UTM zone 43N. The initial experiments used a 512 m square centred near 76.76567 degrees E, 16.30969 degrees N. A separate 128 m square within the image supplied the prototype smoke test. Figure 2 shows both footprints.

Terrain measurements use Copernicus GLO-30 [11], with a window from tile N16E076 and a halo to compute terrain derivatives. GLO-30 is a digital surface model (DSM) and may include vegetation and structures. Its provenance is separate from RGB capture time. One 30 m cell covers an area equivalent to 3600 RGB pixels at the 0.5 m working scale. The Raichur package includes no independently reviewed obstacle-height, restriction, road, or building survey.

The original imagery footprint intersects about 1.54 square kilometres of the 8450.99 square kilometre Raichur district feature used by the project [12]. Both reported runs cover much smaller parts of that image. The prototype test used a district-oriented worker on a local crop.

Historical Bhadra context

Bhadra supplied the existing classifier and an earlier terrain-knowledge example. The historical vision-refined package uses a Sentinel-2 context date of 1 June 2026. Its high-resolution vision imagery has an unknown acquisition date. Its 30 m terrain grid uses FABDEM, which was designed to remove estimated building and tree height biases from Copernicus GLO-30 [13]. Bhadra and Raichur differ in image source, date provenance, geography, and processing policy.

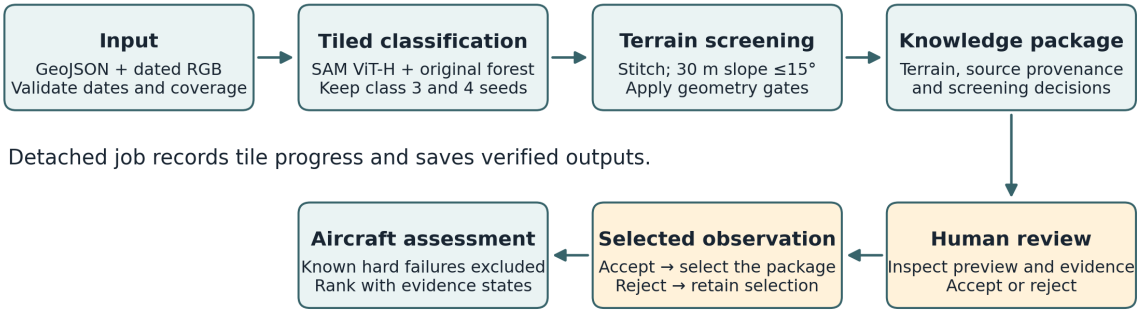


Fig. 1. HELPS workflow. The background job produces a terrain package and audit. The review decision controls which observation is used for aircraft assessment.

IV. WORKFLOW

Observation validation and tile processing

Figure 1 shows the stages. The operator supplies a named region, local capture date, WGS 84 GeoJSON polygon or multipolygon, and a georeferenced 8-bit RGB GeoTIFF. The interactive build limits the region to 4 square kilometres and 8 km across, with a maximum of 32 planned tiles, 256 MiB of RGB input, and 2 MiB of boundary input. The area determines a local UTM projection. The working pixel size is 0.5–2 m, derived from the image or specified by the operator. Input validation rejects coarse sources that would need upsampling into the supported range. Preflight checks footprint and recognized date tags. Per-tile checks verify valid pixels and alpha opacity inside the requested region.

The worker uses cores of up to 896 pixels per side and reads a 64-pixel halo at each edge for model context. It retains only classified geometry in the intersection of a tile core and the region boundary. Each pixel therefore has one core owner, which avoids counting the overlap twice. Labels can still disagree across seams. The job runs separately from the browser and stores completed tile GeoPackages with checksum records. A resume can reuse matching completed outputs.

Mask composition and classification

SAM ViT-H generates automatic mask proposals on each buffered tile. The recorded configuration samples 24 points per side and uses a predicted-IoU threshold of 0.82 and stability threshold of 0.90. These settings filter mask proposals. The GeoSAM composition resolves overlaps, fills uncovered areas using image edges, and merges small fragments. The original forest uses 16 features, including colour and texture. The region rules aggregate the classifier predictions and preserve uncertain labels. In Raichur the road-distance feature was held constant and geographic priors were empty because reviewed local inputs were unavailable.

Let B denote the study boundary, C_j the core of tile j, and P_{ji} a classified polygon with label k_{ji}. The producer dissolves polygons selected from class 3 (crops, grass, and low vegetation) and class 4 (bare soil, rock, and sparse ground) before measuring footprint geometry. This allows pieces split by tile boundaries to be measured together when they form a connected footprint.

$$S = \bigcup (P_{ji} \cap C_j \cap B), \text{for } k_{ji} \in \{3, 4\} \quad (1)$$

Slope and footprint geometry

The worker aligns a covering elevation window to a 30 m metric grid. Horn's 3 × 3 derivative estimates slopes from neighbouring samples [14]. For east and north gradients g_x and g_y, the slope in degrees is given by

$$\theta = (180/\pi) \arctan \sqrt{(g_x^2 + g_y^2)} \quad (2)$$

The derivative calculation includes a one-cell halo and records the area with valid support. The initial seed mask is intersected with the union of valid cells satisfying $\theta \leq 15°$. The resulting pieces require area of at least 900 m^2, both rotated-rectangle dimensions of at least 30 m, and a maximum inscribed clear diameter of at least 30 m. Clear diameter describes the interior of the mapped polygon. Aircraft profiles apply their own thresholds during assessment. The 15 degree threshold is a producer setting.

Review and aircraft assessment

The package stores terrain footprints, measured subregions, and source provenance. A reviewer can inspect the map and source limits before accepting or rejecting the proposal. The regional registry identifies the selected observation. A pending or rejected proposal leaves an existing selection in use.

The assessment engine queries selected terrain within a search radius and uses the configured LCH, LUH, or ALH research profile, mission weights, and available weather and context. Known hard-gate failures exclude an opportunity. Unknown evidence can appear in a research result with the corresponding verification requirements. For supported mission factors with scaled values $f_i \in [0,1]$ and effective weights w_i, the relative priority is given by

$$R = 100 (\Sigma_i w_i f_i) / (\Sigma_i w_i) \quad (3)$$

The engine also reports the coverage of configured weight by the supported factors. An empty set of supported factors yields no score. The priority score has no probabilistic interpretation. An empty shortlist is a valid application outcome.

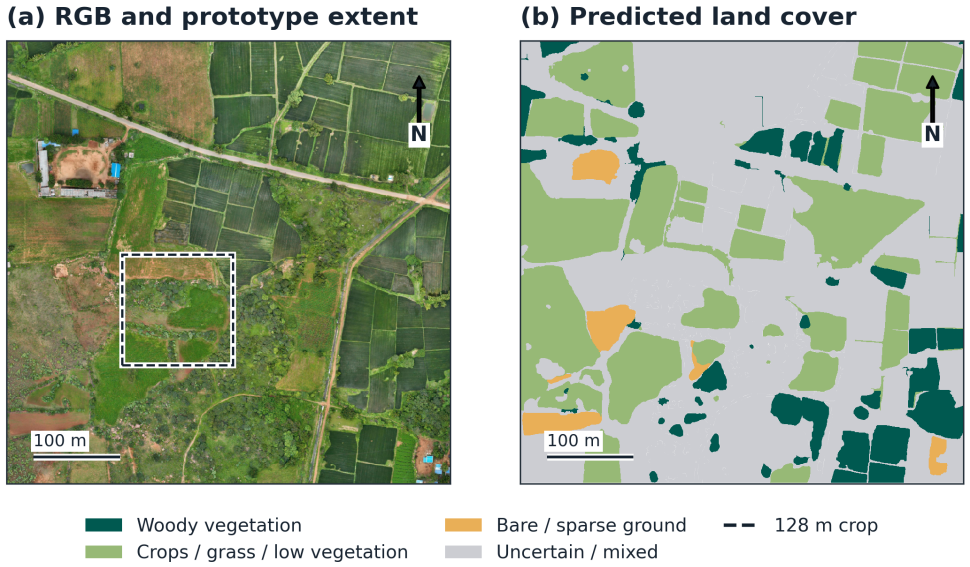


Fig. 2. Raichur historical pilot. (a) The 512 m RGB crop with the 128 m prototype extent dashed. (b) Land-cover predictions from the recovered forest. Imagery by WELL Labs, Open Imagery Network contributor, via OpenAerialMap, CC BY 4.0 [9] [10]. Cropped and resampled.

V. FEASIBILITY RESULTS

512 m historical Raichur pilot

The fixed-segment transfer comparison reused 250 automatic SAM mask proposals to produce 603 classified regions over 262,144 m^2. The recovered Bhadra forest and region rules classified 6,646.75 m^2 as possible bare ground and marked 58.12% of classified area uncertain. Six bare-ground components entered strict screening and failed geometry or clearance. The slope-only layer retained 6,025.50 m^2 at or below 12 degrees in eight fragments. The strict layer contained zero qualifying candidates. The separate forest trained on this same crop marked 16.65% of the classified area uncertain and also yielded no strict candidates. The comparison changed the model and classification rules, with no independently labelled reference set.

A separate integrated legacy build, raichur-rgb-20250901-6c329e7c, regenerated the same mask and region counts. With six CPU PyTorch threads, it recorded 26.71 s for model loading, 142.88 s for SAM inference, and 31.46 s for composition and classification. Its bare-ground producer returned zero candidates. The measurements are specific to this archived run.

128 m Raichur prototype smoke test

The prototype worker processed one tile on the 0.5 m image with both seed classes enabled. It generated 93 proposals on the buffered window, composed 132 regions before clipping, and retained 74 class polygons within the study boundary. Dissolving the seed classes produced nine polygons. The 30 m slope raster covered the full study area. Seven pieces failed geometry or clearance and two qualified (Fig. 3). Table I reports the measured footprints. The largest rejected polygon had 1,168.75 m^2 after clipping but failed the clear-diameter gate. Restriction and obstacle-height evidence remained unknown for both retained polygons. The prototype test exercised terrain qualification for one tile. Aircraft assessment and multi-tile behaviour were outside that test.

TABLE I. Measured terrain footprints in the 128 m prototype crop

ID	Area (m^2)	Width (m)	Clear (m)	Max slope
RGB-1	1736.00	39.00	33.92	10.62°
RGB-5	2127.75	47.61	38.07	11.39°

Integrated build confirmation

The unified application accepted the same 128 m boundary and 0.5 m image through its upload API. The submitting process exited while the detached worker continued. Dataset rgb-rgb-workflow-validation-20250901-6ee09498 reached the review stage with two terrain candidates and five threshold-specific slope subregions, comprising one at 10 degrees, two at 12 degrees, and two at 15 degrees. Multiple slope subregions can refer to the same terrain parent. The repeat uses the prototype input and confirms the integrated upload-to-review path. The study did not measure aircraft ranking performance.

Historical Bhadra package

The earlier Bhadra vision refinement delivered 4474 terrain candidates and 25,931 measured slope subregions. The candidate total includes 1664 fragments of prior candidates and 2810 additions from image-derived regions screened with the historical 12 degree maximum-slope policy. The imagery provenance, elevation source, 5 m contraction, and exclusions differ from the Raichur prototype. These counts describe the historical package. They cannot serve as a benchmark for the new RGB producer.

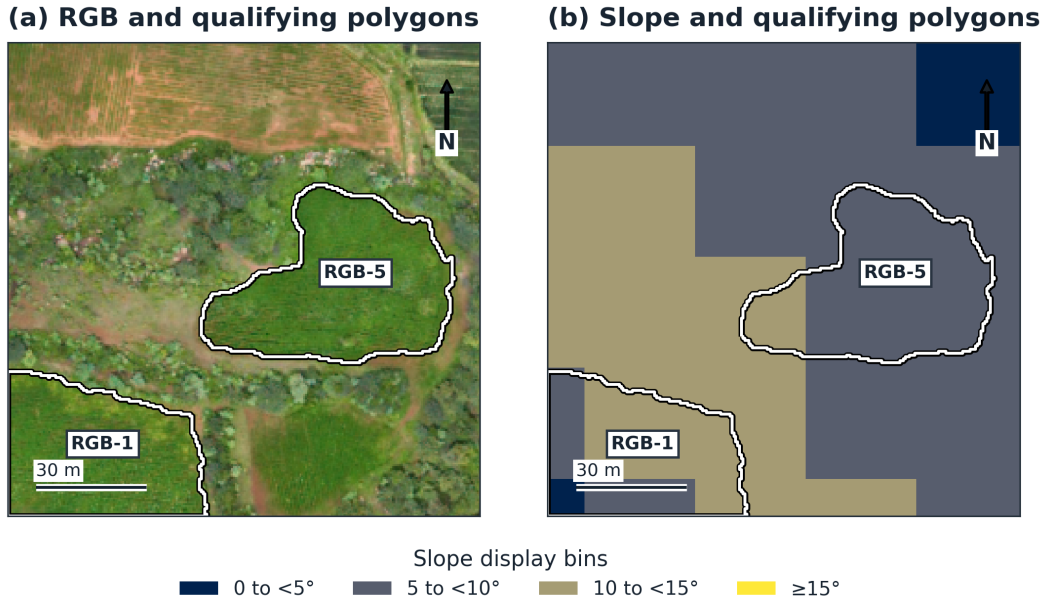


Fig. 3. Raichur prototype screening output. The white outlines show the two footprints that passed the terrain and geometry gates. Panel (a) uses cropped and resampled WELL Labs imagery via OpenAerialMap, CC BY 4.0 [9] [10]. Panel (b) shows slope derived on the 30 m grid. Copernicus DEM copyright DLR e.V. 2010–2014 and Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA. All rights reserved. [11]

VI. EVALUATION DESIGN

The Raichur runs changed multiple variables at once. A controlled study should use held-out tiles sampled across settlement edges, irrigated and dry agricultural land, rocky terrain, woody cover, and relief. Reference annotation should delineate visible openings and hazards independently of the models, retain ambiguous areas, and record inter-reviewer disagreement. Field or fine 3-D evidence would support a separate comparison of local slope, obstacle height, approach geometry, and surface suitability. Olofsson *et al.* provide guidance on sampling and area-adjusted accuracy [15].

For image-scale experiments, resample the same dated orthomosaic and geographic footprint to 0.5, 1, and 2 m with a specified method. Hold elevation, geometry rules, model and seed policy fixed. Report how the ground support of pixel-based texture features and SAM prompt spacing varies with resolution. For a seed-policy ablation, hold the image and all later stages fixed and compare class 4 alone with classes 3 and 4 together. A terrain experiment can compare DSM products on the same image predictions using an independent, finer elevation reference. Seam tests can compare classification overlap and candidate persistence after shifting the tile grid.

Report per-class area-adjusted user and producer accuracy, opening precision and recall at declared polygon intersection-over-union thresholds, overlap with independently annotated visible hazards, empty-shortlist rate, and per-stage time and memory. Use spatial blocks as the resampling unit when estimating uncertainty, and report counts of independent tiles or sites. For the assessment stage, record hard-gate and unknown-state frequencies and rank agreement when rankings exist. Report unmatched opportunities and changes in supported weight coverage when comparing ranks.

VII. LIMITATIONS

Single-date RGB supports visible evidence at the acquisition date. Shadows, crop phenology, transfer of the Bhadra classifier to drone imagery, and missing context can change label assignments. A low-vegetation label leaves height, wetness and bearing capacity unknown. The 30 m DSM cannot consistently resolve wires, ruts, small trees, or local slope variation within a cell. An inscribed disk describes mapped geometry. Approach, departure, wind, aircraft limits, and surface conditions require separate evidence. The Raichur pilots contain no independent field verification.

VIII. CONCLUSION

HELPS converts a dated RGB observation into a documented terrain package whose selection requires review. Historical and prototype Raichur runs exercised the empty and nonempty screening paths. Their audits distinguish missing or unsuitable geometry from coarse slope evidence. Controlled evaluation is needed to determine how image scale and seed policy affect detectable footprints and where additional evidence is necessary for aircraft assessment.

DATA AND REPRODUCIBILITY

The open image source is identified in [9] [10], with elevation terms and attribution in [11]. The project records input and model hashes, settings, tile outputs, screening audits, and package manifests. The pilot evidence is archived under data/experiments in the HELPS checkout. The editable figure generator and reproduction notes accompany this manuscript. No Bhadra basemap imagery is reproduced.

REFERENCES

[1] Federal Aviation Administration, *Helicopter Flying Handbook*, FAA-H-8083-21B, 2019, chs. 5, 10–11. [Online]. Available: https://www.faa.gov/regulations_policies/handbooks_manuals/aviation/helicopter_flying_handbook

[2] S. Scherer, L. Chamberlain, and S. Singh, "Autonomous landing at unprepared sites by a full-scale helicopter," *Robot. Auton. Syst.*, vol. 60, no. 12, pp. 1545–1562, Dec. 2012, doi: 10.1016/j.robot.2012.09.004.

[3] J. Erskine, C. Oxendine, W. Wright, M. O'Banion, and A. Philips, "Evaluating the relationship between data resolution and the accuracy of identified helicopter landing zones (HLZs)," *Appl. Geogr.*, vol. 139, Feb. 2022, Art. no. 102652, doi: 10.1016/j.apgeog.2022.102652.

[4] A. Kirillov *et al.*, "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 4015–4026, doi: 10.1109/ICCV51070.2023.00371.

[5] Q. Wu and L. P. Osco, "samgeo: A Python package for segmenting geospatial data with the Segment Anything Model (SAM)," *J. Open Source Softw.*, vol. 8, no. 89, Sep. 2023, Art. no. 5663, doi: 10.21105/joss.05663.

[6] L. P. Osco *et al.*, "The Segment Anything Model (SAM) for remote sensing applications: From zero to one shot," *Int. J. Appl. Earth Obs. Geoinf.*, vol. 124, 2023, Art. no. 103540, doi: 10.1016/j.jag.2023.103540.

[7] Nehareddy1234, *cv_ai_helps*, GitHub software repository, classifier asset revision 47843a2 and pipeline revision 9e46fe3. [Online]. Available: https://github.com/Nehareddy1234/cv_ai_helps

[8] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, Oct. 2001, doi: 10.1023/A:1010933404324.

[9] WELL Labs, *RAICHUR_D10_2025_09-M1*, OpenAerialMap / Open Imagery Network, item 69493f7684a859b011c94916, Sep. 2025 (local capture date), CC BY 4.0. [Online]. STAC item.

[10] Humanitarian OpenStreetMap Team, *OpenAerialMap*, Open Imagery Network license and attribution terms. [Online]. Available: https://openaerialmap.org/about/

[11] European Union and European Space Agency, *Copernicus Digital Elevation Model, GLO-30*, N16E076. [Online]. Available: https://registry.opendata.aws/copernicus-dem/

[12] D. Runfola *et al.*, "geoBoundaries: A global database of political administrative boundaries," *PLOS ONE*, vol. 15, no. 4, Apr. 2020, Art. no. e0231866, doi: 10.1371/journal.pone.0231866.

[13] L. Hawker *et al.*, "A 30 m global map of elevation with forests and buildings removed," *Environ. Res. Lett.*, vol. 17, no. 2, 2022, Art. no. 024016, doi: 10.1088/1748-9326/ac4d4f.

[14] B. K. P. Horn, "Hill shading and the reflectance map," *Proc. IEEE*, vol. 69, no. 1, pp. 14–47, Jan. 1981, doi: 10.1109/PROC.1981.11918.

[15] P. Olofsson, G. M. Foody, M. Herold, S. V. Stehman, C. E. Woodcock, and M. A. Wulder, "Good practices for estimating area and assessing accuracy of land change," *Remote Sens. Environ.*, vol. 148, pp. 42–57, May 2014, doi: 10.1016/j.rse.2014.02.015.